Hao Hao

Carnegie Mellon University
Heinz College of Information Systems and Public Policy
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RESEARCH INTEREST

Theory: Robust optimization.
Applications: Supply chain management and transportation operations.

EDUCATION

Carnegie Mellon University, Heinz College
Ph.D. in Operations Research — 2021 - 2027

Carnegie Mellon University, College of Engineering
M.S. in Advanced Infrastructure Systems — 2019 - 2021

Imperial College London
M.Eng. (Undergraduate degree)
in Electrical and Electronic Engineering (First-Class Honours) — 2015 - 2019

JOURNAL ARTICLES

Robust Paths: Geometry and Computation
Hao Hao and Peter Zhang
Major Revision in Operations Research
- **First Place, 2025 George Nicholson Student Paper Competition**
- **First Place, 2025 INFORMS Computing Society Student Paper Award**

Assessing the Routing Efficiency of Cities
Hao Hao, Hai Wang and Peter Zhang
To be Submitted to Nature Cities

REFEREED CONFERENCE PROCEEDING

Geometry-Informed Tracing of Model Adaptation Path under Protected-Capability Constraints
Hao Hao and Peter Zhang
NeurIPS 2026 Workshop on ML×OR

WORKING PAPERS

Monotonicity of Robust Resource Allocation Against Uncertainty
Hao Hao, Mark Brennan and Peter Zhang

Submodular Learning for Risk-aware Prediction-Optimization: Applications in Distributed Healthcare
Hao Hao, Holly Wiberg, and Peter Zhang

TEACHING

Teaching Assistant, Heinz College, Carnegie Mellon University
Decision Making under Uncertainty (Graduate-level OR) — 2024, 2025
Decision Analytics for Business and Policy (Graduate-level OR) — 2023, 2024

Unstructured Data Analytics (Graduate-level Data Science) 2024

PROFESSIONAL EXPERIENCE

Deloitte
Optimizing logistics operations for distributed healthcare.
Student Collaborator 2025 - 2026

Bayer AG
Designing resilient supply chains against demand uncertainty and disruptions.
Student Collaborator 2022 - 2023

PRESENTATIONS

Robust Paths: Geometry and Computation
INFORMS Optimization Society Conference 2026
International Conference on Bilevel Optimization 2026
INFORMS Annual Meeting 2025
YinzOR, Carnegie Mellon University 2025
INFORMS Annual Meeting 2024

Assessing and Explaining the Efficiency of Urban Transportation Networks
INFORMS Annual Meeting 2024
INFORMS Annual Meeting 2023
Canadian Operations Research Society Annual Conference 2021

SERVICES

Reviewer: Operations Research, Transportation Science
Session Chair: 2022 INFORMS Annual Meeting; 2026 International Conference on Bilevel Optimization
Organizer: OR PhD Student Seminar, Heinz College, Carnegie Mellon University

AWARDS

First Place, INFORMS George Nicholson Student Paper Competition 2025
The competition is held each year to identify and honor outstanding papers in the field of operations research and the management sciences written by a student.

First Place, INFORMS Computing Society Student Paper Award 2025
The award is given annually to the best paper at the interface of computing and operations research by a student author, as judged by the award selection committee.

George Duncan Award for Excellence in Doctoral Studies for the Second Research Paper, Heinz College, Carnegie Mellon University 2025

TCS Presidential Fellowship at Carnegie Mellon University 2022-2023

ADDITTIONALS

Programming: Python
Interests: Bouldering, running, cooking